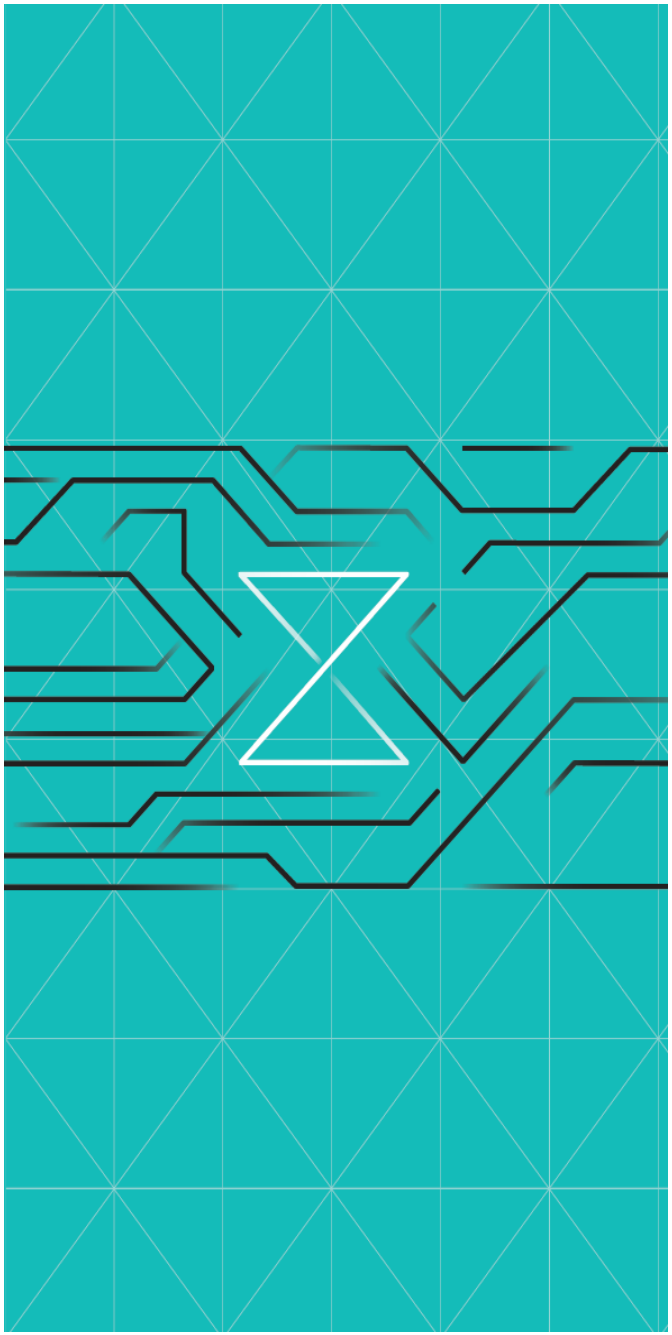


Advanced

250303-1925

LNX2

Linux on Z: Go Worldwide

- TAKE CONTROL OF YOUR LINUX INSTANCE AND START BUILDING
- 1 INSTALL THE WEB SERVER
- 2 APACHE, GO!
- 3 CREATE THE HOME PAGE
- 4 FIND YOUR LINUX IP ADDRESS
- 5 YOU ARE MY FIREWALL
- 6 VIEW YOUR MASTERPIECE
- 7 WATCH THE ACCESS LOG
- 8 WOULD YOU LIKE TO PLAY A GAME?
- 9 UNPACK THE CODE FROM THE TARBALL
- 10 INSTALL BUILD TOOLS
- 11 CONFIGURE AND BUILD
- 12 VERIFY BUILD (PLAY THE GAME)
- 13 NEXT STEPS

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TAKE CONTROL OF YOUR LINUX INSTANCE AND START BUILDING

The Challenge

Linux started as a hobbyist alternative to expensive UNIX systems, which were only available to those in universities or large companies.

Through rapid adoption, it has ended up on servers, laptops, phones, and devices all around the world. Of course, one of those servers is the IBM Z mainframe, where it benefits from the extreme resiliency, performance, and scalability that the platform offers.

Before You Begin

Access to a Linux system is required for this challenge, so you will definitely want to complete the first Linux challenge, **LNX1**, before attempting this challenge.

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Investment

Steps	Duration
15	90 minutes

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1 INSTALL THE WEB SERVER

```
httpd-filesystem.noarch : The basic directory layout for the Apache HTTP server
python38-mod_wsgi.s390x : A WSGI interface for Python web applications in Apache
mod_ldap.s390x : LDAP authentication modules for the Apache HTTP Server
python3-mod_wsgi.s390x : A WSGI interface for Python web applications in Apache
mod_proxy_html.s390x : HTML and XML content filters for the Apache HTTP Server
mod_auth_mellon.s390x : A SAML 2.0 authentication module for the Apache Httpd Server
mod_md.s390x : Certificate provisioning using ACME for the Apache HTTP Server
pcp-export-pcp2spark.s390x : Performance Co-Pilot tools for exporting PCP metrics to Apache Spark
mod_intercept_form_submit.s390x : Apache module to intercept login form submission and run PAM authentication
mod_lookup_identity.s390x : Apache module to retrieve additional information about the authenticated user
 MTM> sudo yum install httpd.s390x
Last metadata expiration check: 0:06:28 ago on Mon 17 Aug 2020 09:30:18 AM EDT.
Dependencies resolved.
================================================================================================================================
 Package                    Architecture    Version                                          Repository                 Size
================================================================================================================================
Installing:
 httpd                      s390x           2.4.37-21.module+el8.2.0+5008+cca404a3           rhel8-appstream           1.3 M
Installing dependencies:
 mailcap                    noarch          2.1.48-3.el8                                     rhel8-base                 39 k
 redhat-logos-httpd         noarch          81.1-1.el8                                       rhel8-base                 26 k
 apr-util                   s390x           1.6.1-6.el8                                      rhel8-appstream           104 k
 apr                        s390x           1.6.3-9.el8                                      rhel8-appstream           122 k
 httpd-filesystem           noarch          2.4.37-21.module+el8.2.0+5008+cca404a3           rhel8-appstream            36 k
 mod_http2                  s390x           1.11.3-3.module+el8.2.0+4377+dc421495            rhel8-appstream           152 k
 httpd-tools                s390x           2.4.37-21.module+el8.2.0+5008+cca404a3           rhel8-appstream           102 k
Installing weak dependencies:
 apr-util-bdb               s390x           1.6.1-6.el8                                      rhel8-appstream            24 k
 apr-util-openssl           s390x           1.6.1-6.el8                                      rhel8-appstream            27 k
Enabling module streams:
 httpd                                      2.4

Transaction Summary
================================================================================================================================
Install  10 Packages

Total download size: 1.9 M
Installed size: 5.6 M
Is this ok [y/N]: y
```

Return to your Linux instance at the LinuxONE Community Cloud (L1CC).

In the ssh terminal, type yum search apache to look for installable packages related to the Apache web server.

httpd.s390x is the one you want. As a note, **s390x** is shorthand for “64 bit Z architecture”. Install that package with the command:

```
sudo yum install httpd.s390x
```

It will show you a number of dependencies that it also needs to install, and ask you if this is OK. Enter ‘y’ to tell it to proceed.

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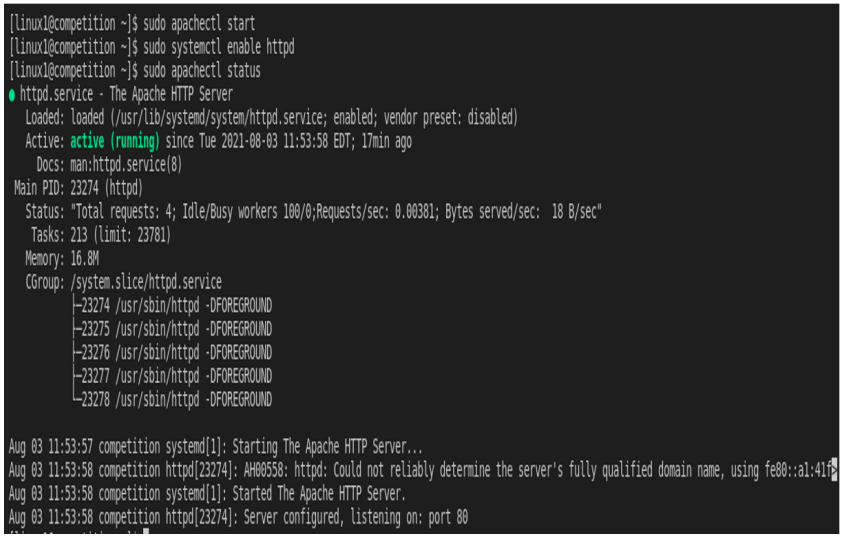
2 APACHE, GO!

You have now installed the Apache web server, but it won't do anything until it is started.

So use the following commands to:

- start it now
- tell the system to enable it for starting automatically
- check on its status (look for "active (running)")

```
sudo apachectl start
sudo systemctl enable httpd
sudo apachectl status
```



If the output of the "status" command is longer than can fit in your window, you may need to scroll to the bottom, or press Q to quit out.

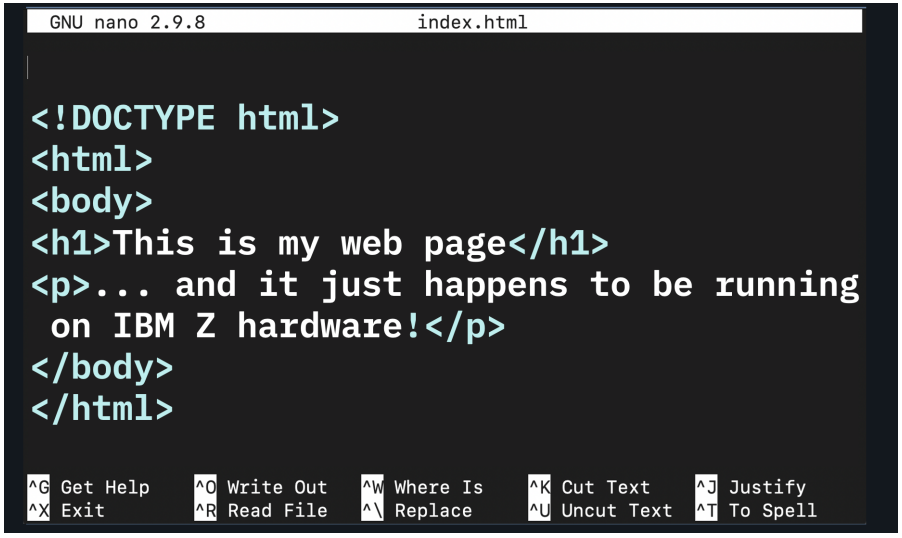
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3 CREATE THE HOME PAGE

Change directory into **/var/www/html** and create a file called “index.html” as the new default home page for the web server.

(Since this is a protected directory, remember that you’ll need root authority to create/save files here...)

```
<!DOCTYPE html>
<html>
<body>
<h1>This is my web page</h1>
<p>... and it just happens to be running
 on IBM Z hardware!</p>
</body>
</html>
```



If you’re familiar with HTML, then go nuts. Otherwise, feel free to copy/paste the sample code above into a nano edit session. When you are ready, press *Control-O* to save and *Control-X* to quit the nano editor.

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(This should not need to be said, but don’t put anything out here in poor taste or questionable judgment. Actions will be taken for improper use of this system.)

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4 FIND YOUR LINUX IP ADDRESS

Before going much further, make sure you know your system's IP address.

It is listed on the LinuxONE Community Cloud page where you created the server instance, or you can run the command

```
ip –c addr
```

(interface address config) and look for the **inet** address attached to the enc1000 interface (shown below with the "3:" prefix).

```
[linux1@competition ~]$ ip -c addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
3: enc1000: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 02:a1:41:5b:85:30 brd ff:ff:ff:ff:ff:ff
    inet 148.100.76.111/22 brd 148.100.79.255 scope global noprefixroute enc1000
       valid_lft forever preferred_lft forever
    inet6 fe80::a1:41ff:fe5b:8530/64 scope link
       valid_lft forever preferred_lft forever
```

Write that down somewhere, or just make a note how you can find it again when you need it (Hint: you *are* going to need it!).

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5 YOU ARE MY FIREWALL

By default, the LinuxONE Community Cloud blocks all incoming access except for SSH.

This is a security precaution to prevent unexpected access or abuse of your system.

`sudo iptables --list`

shows the current state of the firewall rules:

```
[[linux1@rdcl1cc ~]$ sudo iptables --list
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     all  --  localhost            localhost
ACCEPT     all  --  localhost            localhost
ACCEPT     all  --  localhost            localhost
ACCEPT     all  --  localhost            localhost
ACCEPT     tcp  --  anywhere             anywhere             ctstate NEW tcp dpt:ssh
ACCEPT     icmp --  anywhere             anywhere
ACCEPT     all  --  anywhere             anywhere             ctstate RELATED,ESTABLISHED
REJECT     all  --  anywhere             anywhere             reject-with icmp-host-prohibited

Chain FORWARD (policy DROP)
target     prot opt source               destination
```

To look at your new web page, you will need to make an addition in the firewall rules for TCP port 80, on which the web server listens for http requests.

Use the following command to add a rule to iptables to allow http requests on port 80.

```
sudo iptables --insert INPUT --protocol tcp --dport 80 --jump ACCEPT
```

Note that each of the command options is preceded by two "-" characters, not one

Check that the rule has been applied:

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`sudo iptables --list`

and you should now see:

```
[[linux1@rdcl1cc html]$ sudo iptables --list
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:http
ACCEPT     all  --  localhost            localhost
ACCEPT     all  --  localhost            localhost
ACCEPT     all  --  localhost            localhost
ACCEPT     all  --  localhost            localhost
ACCEPT     tcp  --  anywhere             anywhere             ctstate NEW tcp dpt:ssh
ACCEPT     icmp --  anywhere             anywhere
ACCEPT     all  --  anywhere             anywhere             ctstate RELATED,ESTABLISHED
REJECT     all  --  anywhere             anywhere             reject-with icmp-host-prohibited

Chain FORWARD (policy DROP)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
```

(The `dport` (destination port) is shown here as `dpt:http` - indicating that port 80 has been recognized as the standard port for http server)

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iptables is “live”

The commands to the iptables firewall take effect **immediately**

Take great care if you try other iptables commands;if you delete the rule that enables ssh, you will not be able to login to the virtual machine again.

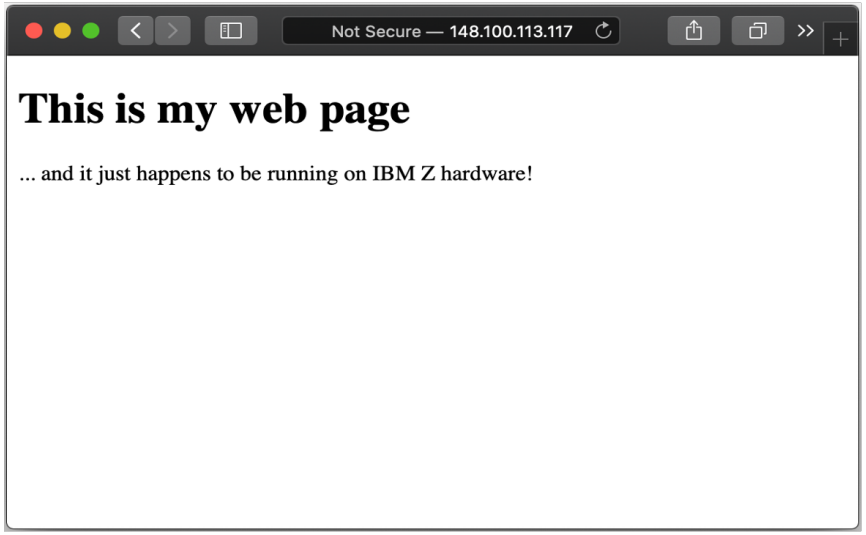
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6 VIEW YOUR MASTERPIECE

Open a web browser and connect to the IP address of your system.

For example, http://148.100.244.250

If the server is running, the firewall is modified and enabled, and you saved the HTML in *index.html*, you should be able to see your creation in a web browser.



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7 WATCH THE ACCESS LOG

Do you remember using the tail command to watch a file as it changes?

Enter the command:

```
sudo tail -f /var/log/httpd/access_log
```

This file is a record of any time your web page has been requested, including the date+time, the requesting IP address, and browser information.



Watch the access log update as you request your web page from a browser window.

If you can, try to open it from another browser on another computer or a phone and see the differences.

Press *Control-C* to end the tail command, when you have seen enough.

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“WHY DO I HAVE TO KEEP TYPING SUDO?”

Perhaps you have heard the expression “Measure twice, cut once”. The idea there being that if you rush during the planning stages, you are more likely to make a mistake.

If only you had taken another few seconds to think about what you were about to do, it could have saved a lot of time and heartache.

By using a default non-privileged user, and only elevating your access when you *really* need it, you are less likely to accidentally erase an important file or directory. It reminds you that you are about to do something that could have potentially dangerous consequences.

Whenever you are about to use a command with *sudo*, take a few seconds to double-check your command *before* you press Enter.

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8 WOULD YOU LIKE TO PLAY A GAME?

Sometimes the software you want isn't readily available through a package manager like *yum*, or in the spirit of open source software, you want to make your own adjustments to the source code and build things on your own.

That is all possible with the compilers available here.

In the next few steps, you will

- download the public source code for a game, **Vitetris**
- extract the source code
- compile that code so it will run on your system
- play the game!

Curl is a command-line utility for interacting with URLs. It supports FTP, HTTP, LDAP, and many other protocols. In this case, we're going to use it to download a file from a remote web server onto our Linux server.

Download the game source code with the following command:

```
curl -LO https://github.com/vicgeralds/vitetris/archive/v0.58.0.tar.gz
```

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```
 <ZXP> curl -LO https://github.com/vicgeralds/vitetris/archive/v0.58.0.tar.g
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Curr
                                 Dload  Upload   Total   Spent    Left  Spee
100   128  100   128    0     0   2666      0 --:--:-- --:--:-- --:--:--  2
100 96957    0 96957    0     0   422k      0 --:--:-- --:--:-- --:--:--  5
 <ZXP> ls -lah
total 116K
drwx------. 5 linux1 linux1 168 Aug 17 11:02 .
drwxr-xr-x. 3 root   root    20 Jul 27 16:33 ..
-rw-------. 1 linux1 linux1 482 Aug 14 12:49 .bash_history
-rw-r--r--. 1 linux1 linux1  18 Aug 30  2019 .bash_logout
-rw-r--r--. 1 linux1 linux1 141 Aug 30  2019 .bash_profile
-rw-r--r--. 1 linux1 linux1 312 Aug 30  2019 .bashrc
drwx------. 3 linux1 linux1  20 Jul 27 16:47 .config
-rw-------. 1 linux1 linux1  35 Aug 14 11:44 .lesshst
drwxrwxr-x. 2 linux1 linux1  26 Aug 12 12:03 my_directory
drwx------. 2 linux1 linux1  29 Jul 27 16:33 .ssh
-rw-rw-r--. 1 linux1 linux1 95K Aug 17 11:02 v0.58.0.tar.gz
 <ZXP>
```


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9 UNPACK THE CODE FROM THE TARBALL

The file you just downloaded has an odd suffix, in fact it has got two; **tar** and **gz**.

This tells you that it has been both encoded and compressed so that its structure and permissions will be preserved after uploading and downloading.

Type the following command to extract the file, which is commonly referred to as a “tarball”

`tar xvf v0` (then just hit tab to auto-complete the rest)

then press Enter.

```
drwx------. 2 linux1 linux1  29 Jul 27 16:33 .ssh
-rw-rw-r--. 1 linux1 linux1 95K Aug 17 11:02 v0.58.0.tar.gz
[<ZXP> tar xvf v0.58.0.tar.gz
vitetris-0.58.0/
vitetris-0.58.0/.gitignore
vitetris-0.58.0/INSTALL
vitetris-0.58.0/Makefile
vitetris-0.58.0/README
vitetris-0.58.0/changes.txt
vitetris-0.58.0/config.mk
vitetris-0.58.0/configure
vitetris-0.58.0/icon.ico
vitetris-0.58.0/icon.rc
vitetris-0.58.0/licence.txt
vitetris-0.58.0/pc8x16.fnt
vitetris-0.58.0/src-conf.sh
vitetris-0.58.0/src/
vitetris-0.58.0/src/Makefile
vitetris-0.58.0/src/cfgfile.c
vitetris-0.58.0/src/cfgfile.h
```

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10 INSTALL BUILD TOOLS

Before we can go much further, we need to install some development tools. Use *yum* to install **make** and **gcc**

make is a utility for building software from a configuration file, and **gcc** is the GNU C Compiler, used to compile software from source.

To build the game from source code, you will need both.

```
<ZXP> sudo yum install make gcc
Last metadata expiration check: 0:02:25 ago on Mon 17 Aug 2020 11:23:02 AM EDT.
Dependencies resolved.
================================================================================
 Package              Arch     Version                   Repository         Size
================================================================================
Installing:
 make                 s390x    1:4.2.1-10.el8            rhel8-base        496 k
 gcc                  s390x    8.3.1-5.el8               rhel8-appstream    18 M
Installing dependencies:
 pkgconf-pkg-config   s390x    1.4.2-1.el8               rhel8-base         15 k
 libxcrypt-devel      s390x    4.1.1-4.el8               rhel8-base         25 k
 pkgconf              s390x    1.4.2-1.el8               rhel8-base         38 k
 pkgconf-m4           noarch   1.4.2-1.el8               rhel8-base         17 k
 libpkgconf           s390x    1.4.2-1.el8               rhel8-base         34 k
 glibc-headers        s390x    2.28-72.el8               rhel8-base        461 k
 glibc-devel          s390x    2.28-72.el8               rhel8-base        1.0 M
 libatomic            s390x    8.3.1-5.el8               rhel8-base         21 k
 libubsan             s390x    8.3.1-5.el8               rhel8-base        143 k
 binutils             s390x    2.30-73.el8               rhel8-base        5.9 M
 libgomp              s390x    8.3.1-5.el8               rhel8-base        198 k
 libasan              s390x    8.3.1-5.el8               rhel8-base        370 k
 kernel-headers       s390x    4.18.0-193.14.3.el8_2     rhel8-base        4.0 M
 isl                  s390x    0.16.1-6.el8              rhel8-appstream   801 k
```

Some Linux distributions include these types of development tools by default. Other times, it makes sense to leave them out to save disk space. In this case, you will have to install them, but it should only take a few seconds.

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11 CONFIGURE AND BUILD

Change directory into the **vitetris*** directory and type `./configure` to set everything up, and then `make` to actually build the program from source.

Make sure you include the dot-slash before *configure* or else it probably not work.

```
<ZXP> pwd
/home/linux1/vitetris-0.58.0
<ZXP> ./configure && make
checking for linux/joystick.h... yes
checking for linux/input.h... yes
checking for Xlib... no
checking for conio.h... no
checking for stdint.h... yes
checking for sys/types.h... yes
checking for sys/select.h... yes
checking for sys/un.h... yes
updating src/config.h
updating config.mk

INSTALLATION DIRECTORIES:
  prefix = /usr/local
  bindir = $(prefix)/bin
  datarootdir = $(prefix)/share
  docdir      = $(datarootdir)/doc/vitetris
  pixmapdir   = $(datarootdir)/pixmaps
  desktopdir = $(datarootdir)/applications
```

You can also chain the command together with a pair of ampersands (&&), as in the screenshot. This means “Run this command, and if it runs without errors, then run this command, too”. It will spin for a little bit, and then let you know when it is finished.

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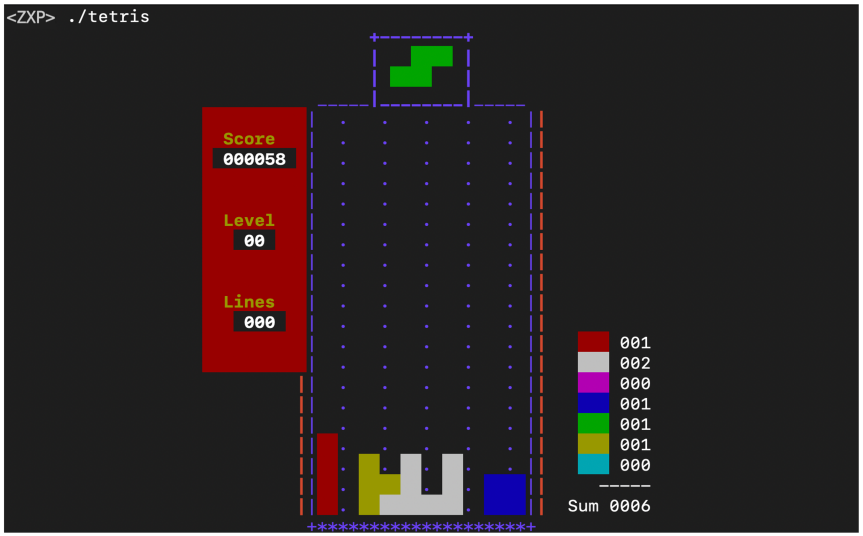
12 VERIFY BUILD (PLAY THE GAME)

It seems silly to go through all of the trouble of downloading and configuring a game and then NOT play it, right?

Type

`./tetris`

and it's "Game: On!"



We could include a lot more text here to help fill up that empty space at the bottom, but you're playing Tetris on a mainframe, so we'll just end there :-)

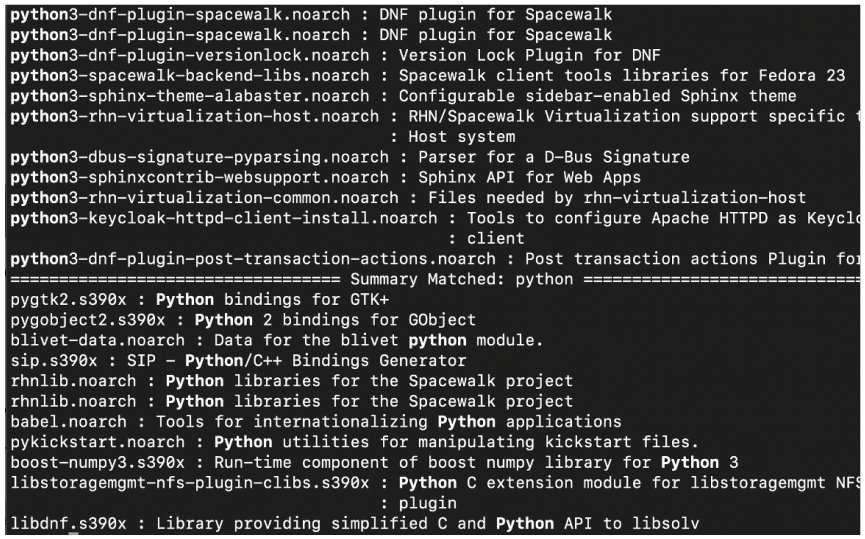
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13 NEXT STEPS

Where you go from here is up to you.

You can keep exploring Linux and use this as your go-to system, or maybe keep installing software (see python examples below) or build out that fine web page.



If you intend to ”Master the Mainframe”, having Linux system skills can be mighty handy.

You mark this challenge completed by running another *curl* command:

```
curl -X POST -d email={your-ibmzxplore-email} http://192.86.32.12:1880/LNX2
```

Make sure to include the email address parameter.

Note that this command uses the **POST** option - the email parameter is sent as the “body” of the request, rather than as a URL option

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Nice job - let's recap	Next up ...
You may have been asking yourself what any of this has to do with enterprise systems... or even mainframe in general. As you may know, an IBM Z mainframe is able to run many different operating systems at the same time. One of those operating systems is z/OS and it also happens to excel at running Linux. In most big companies, there is a mix of z/OS, Linux, and possibly other operating systems (zVM, zVSE, zTPF) being used. This is Linux on IBM Z hardware, so keep that in mind.	Where to go from here? Well, let your interest be your guide. If you want to do more coding/compiling, look into software sources for tools you might like to experiment with. Maybe you want to look into those "stacks" mentioned earlier. Wherever you end up, make sure you have fun!

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